

AZIZ, Karim

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EDUCATION

Boston University, Boston, MA, U.S.A.

September 2023 - May 2027 (Expected)

Bachelor of Arts in Mathematics (Pure Track), Bachelor of Arts in Physics (Graduate Option)

- Grade Point Average: 3.89/4.00
- Relevant Coursework: Honors Vector Calculus, Honors Differential Equations, Honors Linear Algebra, Discrete Mathematics, Abstract Algebra I & II, Real Analysis I & II, Measure Theory, Topology, Algebraic Geometry, Differential Geometry, Algebraic Number Theory, Representation Theory, Classical Mechanics, Modern Physics, Electrodynamics I & II, Quantum Physics I & II, Introduction to Computer Science, Research Writing
- Did an exchange with the University of Geneva in the spring semester of 2026

Collège de la Sainte Famille (Jesuits), Cairo, Egypt

September 2019 - July 2023

French Baccalauréat with Mathematics and Physics-Chemistry Specialty

- Grade Average: 18.26/20.00
- Relevant Coursework: Mathematics, Physics, Chemistry, Programming, Elementary Statistics

EXPERIENCE

European Organization for Nuclear Research (CERN), Geneva, CH

January 2026 - Present

Visiting Researcher, Supervisor: Prof. Alba Grassi

- Research project in mathematical physics and gauge theory.
- Proved group-theoretic identities relating analytic square-integrable solutions of a deformation of the quantum harmonic oscillator to the Quantum Toda Lattice, an integrable system.
- Investigating the convergence of functions in supersymmetric gauge theory, and studying their combinatorial properties in certain limits (Nekrasov–Shatashvili free energy).

Boston University, Boston, MA, U.S.A.

April 2025 - Present

Undergraduate Researcher, Supervisor: Prof. Anatoli Polkovnikov

- Investigated an effective model for classical many-body systems modeling phonon-electron interactions.
- Solved for analytic solutions of these systems and wrote computational simulations to verify them.
- Currently working on solving systems with increasingly larger numbers of vibrational modes.

Boston University, Boston, MA, U.S.A.

June - August 2025

Research Experience for Undergraduates “Ambassador”

- Supervised a group of students who did NSF-funded research at Boston University while helping them discover what Boston has to offer, whether academic or cultural.

RESEARCH

American Physical Society March Meeting, Denver, CO, U.S.A.
Oral Presentation

March 2026

- Presented research conducted under Prof. Polkovnikov at APS's annual conference.

Boston University, Boston, MA, U.S.A.
Poster Presentation

August 2025

- Presented a poster on research conducted over the summer under Prof. Polkovnikov.

GRANTS AND SCHOLARSHIPS

Boston University, Boston, MA, U.S.A.
Full-Ride "University Scholarship"

September 2023 - Present

- Received a scholarship which covers four years of tuition and living expenses. This scholarship is awarded to students "with exceptional financial need and outstanding academic achievement and promise."

European Organization for Nuclear Research (CERN), Geneva, CH
United States Department of Education Grant

July 2026 - August 2026

- Received funding to conduct full-time research with Prof. Alba Grassi in Geneva.

Boston University, Boston, MA, U.S.A.
National Science Foundation (NSF) Grant Number PHY-2244795

June - August 2025

- Conducted research under Prof. Polkovnikov through the Research Experience for Undergraduates.

SKILLS

Computational: Python | Unix | LaTeX | Wolfram Mathematica

Linguistic: English (*Native*) | French (*Native*) | Arabic (*Native*) | Spanish (*Fluent*) | German (*Elementary*)

Extracurricular: Classical Music (*Pianist, Composer*) | Chess (*FIDE Profile*) | Soccer Player (*Right Forward*)